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Transition Metal Transformations

Abstract: The purpose of this experiment is to illustrate the effects of structural change on transition metal complexes and their colors. A structural change can mean simply a change in geometry, or an actual change in chemical substituents. The structure of transition metal complexes are greatly influenced by the available ligands in the surroundings. A ligand in this case, is an ion or molecule, usually quite polar and/or charged, that can form a coordinate covalent bond with a transition metal to form a complex. In the biological sciences, a ligand can also be the substrate for an enzyme. Similarly, in both scenarios, the binding ability of the ligand depends greatly on the chemical properties of both the ligand and the solvent.

Safety Precautions: Safety goggles and plastic gloves are required when performing this experiment. Barium chloride is toxic. Click here for MSDS . 1-butanol is flammable and should be kept away from open flame. Avoid prolonged contact and/or inhalation whenever using chemicals.

Experiment 1:

Materials:
copper(II) chloride
concentrated HCl or NaCl
distilled water
magnetic stirrer or glass stirring rod
1-butanol

Procedure A:

1. Prepare a .2M solution of Copper(II) chloride to 100 mL. To do so add 3.41 g. to 100 mL water.
2. Place on magnetic stirrer and put in magnet.
3. Add concentrated NaCl or HCl in small increments while stirring. Be sure to note how much is being used.
4. Stop adding when a change in color is noticed and record amount of additive used.

NaCl can be used as a solid salt, or as a saturated solution. If using a solution of salt, be sure to note the starting and finishing concentrations. Slight heating may aide in dissolving the salt.

Procedure B:

1. Dissolve about 1 g of Copper(II) chloride in about 30 mL of 1-butanol. The solution should be green.
2. Add about 30 mL of water and mix.
3. Stop mixing and allow layers to separate. Note the color of each layer. The lower aqueous layer should be blue, while the upper butanol layer should remain green.
4. HCl or NaCl can be added until both layers turn green.

To view a short movie of the color change click Naclspin.mov

Further Experiments can be done by simply replacing the NaCl with a different ligand. Try ammonium (NH_4^+), potassium iodide (KI), sodium hydroxide (NaOH), potassium thiocyanate (KSCN), etc. Which parts of these compounds is the ligand? Is it a cation, or an anion? Is it positively or negatively charged? Is there a general order of ligand field strengths?

Experiment 2:

Safety Precautions: Safety goggles and plastic gloves are required when performing this experiment. Barium chloride is toxic. Click here for MSDS .

Materials:
8 100 mL beakers
iron(II) ammonium sulfate solution
  made by adding 5-10 g of solid per liter of distilled water (concentration is not important and required volume is approximately 100 mL per run)
iron (III) ammonium sulfate solution
  made by adding 5-10 g of solid per liter of distilled water
400 mL of distilled water
  50 mL per beaker
4 g potassium ferricyanide ($K_3Fe(CN)_6$)
4 g barium chloride ($BaCl_2$)

4 g sodium hydrogen sulfite ($NaHSO_3$)
4 g potassium thiocyanate (KSCN)

Procedure:

1. Add 50 mL of distilled water to each beaker.
2. Add approximately 2 g of each powder to a separate beaker. There should be two beakers for each powder.
3. Add the iron(II) ammonium sulfate solution to one set of four beakers until the solutions change color. This should not require more than 10 mL of iron solution. The reactions should result in a change from all clear solutions to a distinct color:
   - -deep green with crystals for potassium ferricyanide,
   - -opaque off-white for barium chloride,
   - -yellow for sodium hydrogen sulfite,
   - -light red, pink, or clear for potassium thiocyanate.
4. Add the iron (III) ammonium sulfate to the remaining set of four beakers until a color change is observed. Again, this should only require 10 mL of iron solution. The reaction should yield:
   - -light green or clear for potassium ferricyanide,
   - -opaque white for barium chloride,
   - -yellow orange for sodium hydrogen sulfite,
   - -deep red for potassium thiocyanate.
5. For a classroom observation of the different colors, the solutions can be poured into petri dishes and displayed through an overhead projector.

Discussion: Transition metal complexes have geometries based on their ligand fields. They can be either octahedral or tetrahedral based on the splitting of the d orbitals. The strength of the ligands determines its shape and depends on the ability of the ligand to form coordinate covalent bonds with the metal ion. The solution changes from the species $Cu(H_2O)_6^{2+}$ or $[Cu(H_2O)_5Cl]^+$ to the species $CuCl_4^{2-}$upon addition of Cl^-. The former has six ligands in the field, whereas the latter has four, giving a change in configuration from octahedral to tetrahedral. The Cl^- ions, when at a sufficient concentration will completely replace the water in the field of the copper, altering the energy of the d orbitals, and thus, changing the confirmation and its light absorption. This replacement occurs in a stepwise fashion, with the addition of one Cl^- ion at a time. The addition of each ion has its own equilibrium constant k (k=[products]/[reactants], concentrations are in mol/L). An aprotic (non polar) solvent stabilizes the latter species due to the lack of a stronger binding ligand (i.e. water). Water would induce the former more readily, and thus, requires the addition of Cl^- ions at high concentration for the transition to occur. The assembly of coordinate covalent bonds provides metal complexes with an ability to absorb energy in the form of light. The different geometries differ in the arrangement and number of the bonds, accounting for differing absorption abilities, and thus different colors. Iron(II) and iron(III) are used to demonstrate the effects of oxidation states on a reaction. The color changes represent the changes in configurations and the interactions between the various compounds. By using to two different oxidation states of iron it is shown how the presence of a single additional electron can effect the molecular geometry, and thus the resulting products and their color.

1. Why is there a color change? (Think on a molecular level)
2. What ways of changing a solutions color are there? Which of these different ways does this experiment represent?
3. Which reaction yields the same color, exact in shade and consistency, for both iron solutions? Why is it different than the rest of the reactions?

The absorption spectra of the solutions can be determined to reveal the energies at which they most absorb. To perform such an analysis, refer to the lab on spectral analysis within this site.

References:

1. Nassau, Kurt; (1983) *The Physics and Chemistry of Color: The Fifteen Causes of Color* , John Wiley & Sons, Inc; p77-105

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3. Summerlin, L. R. & amp; Ealy, J. L. (1988). *Chemical demonstrations: A sourcebook for teachers. Volume 1* . Washington, D. C.: American Chemical Society